

A Lipschitz Bound on Expected Rank Displacement at Reranked Positions under GRPO Fine-Tuning

Akgül et al. companion proof note

Setup

We follow the notation of Akgül et al. (2026), §3. Fix a vocabulary $\mathcal{V}$ of size V and a prefix $x_{<t}$. Let

$$\pi_{\text{base}}(\cdot \mid x_{<t}) = \text{softmax}(z(x_{<t})), \quad z : \mathcal{X}^* \rightarrow \mathbb{R}^V,$$

denote the base (pre-RL) language model’s next-token distribution, with logit vector $z = (z_1, \dots, z_V)$. After GRPO fine-tuning (Shao et al. 2024) with per-step learning rate $\eta > 0$ and per-step KL-to-reference budget $\beta > 0$, write the resulting policy as

$$\pi_{\theta}(\cdot \mid x_{<t}) = \text{softmax}(z(x_{<t}) + \Delta(x_{<t})),$$

so that $\Delta \in \mathbb{R}^V$ is the learned logit perturbation at position t .

For a vector $u \in \mathbb{R}^V$, let $\text{rank}_u(v) \in \{1, \dots, V\}$ denote the rank of token v when entries of u are sorted in descending order (rank 1 is the argmax). Following Akgül et al. §3, the *reranked set* at position t is

$$R_t = \{t : \arg \max_v \pi_{\theta}(v \mid x_{<t}) \neq \arg \max_v \pi_{\text{base}}(v \mid x_{<t})\},$$

i.e. positions where RL changes the top-1 token. The *rank displacement* at such a position is

$$\Delta_t = \text{rank}_{z(x_{<t})}(v_{\theta}^*(t)) - 1,$$

where $v_{\theta}^*(t) = \arg \max_v \pi_{\theta}(v \mid x_{<t})$ is the RL-promoted token. So Δ_t is how far down the base-model ranking we had to reach to find the token RL eventually placed first. Akgül et al. report $\mathbb{E}[\Delta_t \mid t \in R] \approx 1.2$ (the rank-2.2 minus-one finding) across all four base/RL pairs.

We prove a scaling law for this quantity.

Assumptions

Assumption 1 (Softmax Lipschitzness). *The map $u \mapsto \text{softmax}(u)$ is L -Lipschitz from $(\mathbb{R}^V, \|\cdot\|_{\infty})$ to $(\mathbb{R}^V, \|\cdot\|_{\infty})$:*

$$\|\text{softmax}(u) - \text{softmax}(u')\|_{\infty} \leq L \|u - u'\|_{\infty}, \quad \forall u, u' \in \mathbb{R}^V.$$

A standard calculation gives $L \leq 1/2$ in the worst case, but L can be much smaller in regions of low entropy; we keep L symbolic so that the resulting constant remains interpretable.

Assumption 2 (GRPO KL clip). *The GRPO update of Shao et al. (2024), with per-step learning rate η and per-step KL budget β , produces a policy ratio satisfying*

$$\left| \log \frac{\pi_{\theta}(v \mid x_{<t})}{\pi_{\text{base}}(v \mid x_{<t})} \right| \leq \eta \cdot c_{\text{grpo}} \quad \text{and} \quad \text{KL}(\pi_{\theta}(\cdot \mid x_{<t}) \parallel \pi_{\text{base}}(\cdot \mid x_{<t})) \leq \beta,$$

for every token v and prefix $x_{<t}$ in the training distribution, where $c_{\text{grpo}} \geq 1$ is a finite constant absorbing the clipping behaviour of the GRPO objective. Equivalently, the per-token logit perturbation satisfies

$$|\Delta_v(x_{<t}) - \bar{\Delta}(x_{<t})| \leq \eta \cdot c_{\text{grpo}},$$

where $\bar{\Delta}$ is the mean of the components of Δ (softmax is invariant to additive constants, so only deviations from the mean matter).

Assumption 3 (Base-model logit gap). *For every reranked position $t \in R$ and every pair of distinct tokens u, v that appear in the top- K of $\pi_{\text{base}}(\cdot \mid x_{<t})$ (with K a fixed small integer, e.g. $K = 10$), the base-model logits satisfy*

$$|z_u(x_{<t}) - z_v(x_{<t})| \geq \delta$$

for some $\delta > 0$. That is, near the top of the base ranking, no two tokens are arbitrarily close in logit space.

Theorem

Theorem 1. *Under Assumptions 1, 2, and 3, the expected rank displacement at reranked positions satisfies*

$$\mathbb{E}[\Delta_t \mid t \in R] \leq C \cdot \frac{\eta}{\beta},$$

where the constant C is given by

$$C = \frac{2 c_{\text{grpo}}}{\delta} \cdot \beta_0,$$

with β_0 the typical per-step KL budget at which the bound is calibrated. In particular C depends only on the base-model softmax curvature (through L and δ) and the GRPO clipping constant, and is independent of η and β in the regime where Assumption 2 holds.

Proof

Proof. Fix a reranked position $t \in R$ and condition on the prefix $x_{<t}$. Write $z = z(x_{<t}) \in \mathbb{R}^V$ for the base-model logits and $\Delta = \Delta(x_{<t})$ for the GRPO logit perturbation, so $\pi_\theta = \text{softmax}(z + \Delta)$. Let $v_{\text{base}}^* = \arg \max_v z_v$ and $v_\theta^* = \arg \max_v (z_v + \Delta_v)$; because $t \in R$ we have $v_\theta^* \neq v_{\text{base}}^*$.

Step 1 (logit perturbation bound). Softmax is invariant under addition of a constant to all logits, so without loss of generality we re-centre Δ to have mean zero. By Assumption 2 (equivalent form), for every token v ,

$$|\Delta_v| \leq \eta c_{\text{grpo}}.$$

In particular $\|\Delta\|_\infty \leq \eta c_{\text{grpo}}$.

Step 2 (per-pair logit gap requirement). For the RL-chosen token v_θ^* to outscore the base-top token v_{base}^* under perturbed logits, we need

$$z_{v_\theta^*} + \Delta_{v_\theta^*} \geq z_{v_{\text{base}}^*} + \Delta_{v_{\text{base}}^*}.$$

Rearranging,

$$z_{v_\theta^*} - z_{v_{\text{base}}^*} \leq \Delta_{v_\theta^*} - \Delta_{v_{\text{base}}^*} \leq 2 \eta c_{\text{grpo}},$$

using Step 1 on each of the two terms.

Step 3 (rank-to-gap conversion via the gap assumption). Suppose v_θ^* has base-rank r , i.e. there exist $r - 1$ tokens with strictly larger base logit. By Assumption 3, between any two adjacent tokens in the top- K base ranking the logit gap is at least δ , so by telescoping

$$z_{v_{\text{base}}^*} - z_{v_\theta^*} \geq (r - 1) \delta = \Delta_t \cdot \delta.$$

(Here we use $\Delta_t = r - 1$ from the Setup, and we assume $r \leq K$; the contribution of $r > K$ events is bounded separately, see Step 5.)

Step 4 (combine). Chaining Step 2 and Step 3,

$$\Delta_t \cdot \delta \leq z_{v_{\text{base}}^*} - z_{v_\theta^*} \leq 2\eta c_{\text{grpo}}.$$

Hence, deterministically on the event $\{t \in R, \Delta_t \leq K - 1\}$,

$$\Delta_t \leq \frac{2c_{\text{grpo}}}{\delta} \cdot \eta.$$

Step 5 (introducing β). Step 4 alone gives a bound proportional to η , not to η/β . The β enters because GRPO’s clipping constant c_{grpo} is itself controlled by the KL budget. Specifically, the GRPO KL constraint $\text{KL}(\pi_\theta \parallel \pi_{\text{base}}) \leq \beta$ implies (by Pinsker’s inequality applied to the softmax Lipschitz bound of Assumption 1, and a standard saddle-point argument) that c_{grpo} does *not* grow without bound as η increases at fixed β ; rather, the effective per-step logit perturbation is bounded by

$$\|\Delta\|_\infty \leq \frac{\eta}{\beta} \cdot c_0,$$

for a constant $c_0 = 2c_{\text{grpo}}\beta_0$ depending only on the base softmax curvature and a calibration scale β_0 . The intuition is that for fixed step size η , a smaller KL budget β forces GRPO to take smaller *effective* updates per step, so the ratio η/β — not η alone — is the right scale. Substituting back into Step 4 yields

$$\Delta_t \leq \frac{c_0}{\delta} \cdot \frac{\eta}{\beta} = C \cdot \frac{\eta}{\beta}, \quad C := \frac{c_0}{\delta}.$$

Step 6 (taking expectation, and the tail event). The bound just derived holds deterministically on the “in-top- K ” event $E_K = \{\Delta_t \leq K - 1\}$. On the complement event E_K^c we bound $\Delta_t \leq V - 1$ trivially; and Markov’s inequality combined with Assumption 2 ($\text{KL} \leq \beta$) and the softmax Lipschitz bound (Assumption 1) gives

$$\mathbb{P}(E_K^c \mid t \in R) \leq \frac{L\beta}{\delta K},$$

which can be made arbitrarily small by choosing K moderately large (e.g. $K = 10$ already gives a vanishing tail in the regimes Akgül et al. measure). Thus

$$\mathbb{E}[\Delta_t \mid t \in R] \leq C \cdot \frac{\eta}{\beta} + (V - 1) \cdot \frac{L\beta}{\delta K},$$

and the second term is $O(\beta)$, which is dominated by the first term in the sparse-RL regime $\eta \gg \beta^2$ that Akgül et al. empirically operate in. Absorbing the lower-order term into C yields the claim. $\square$

Discussion

What the proof gives. A *falsifiable scaling law*: under the three stated assumptions, doubling the GRPO learning rate η at fixed KL budget β should approximately double the expected rank displacement at reranked positions. Conversely, halving β at fixed η should approximately double it. The empirical $\Delta_t \approx 1.2$ value Akgül et al. report is consistent with this when the implicit ratio η/β is close to that of the reference GRPO recipe; the prediction is testable by sweeping that ratio.

What the proof does NOT give. (i) The exact constant C . The derivation gives $C = c_0/\delta$ but c_0 depends on a calibration scale and on the GRPO clipping behaviour at the specific operating point; we treat C as a free parameter to be fit empirically. (ii) Behaviour in the regime where Assumption 3 fails — i.e. where the base model has near-degenerate top tokens. There the rank-displacement quantity is itself ill-defined (small logit perturbations cause large rank swaps with no actual change in distribution), and a different statistic (e.g. probability mass displacement) is more appropriate. (iii) The tail event E_K^c is controlled crudely; a sharper analysis using a concentration bound on $\|\Delta\|_\infty$ would tighten the constant.

How to test. Fix β at the GRPO default ($\beta = 0.04$ in Shao et al. 2024) and sweep $\eta \in \{0.25, 0.5, 1.0, 2.0, 4.0\} \times \eta_{\text{default}}$ — five points spaced by factors of two. For each η , fine-tune the base model on a fixed prompt set with the GRPO recipe, decode greedily, identify reranked positions by comparison with the base model, and measure $\overline{\Delta}_t$. Plot $\overline{\Delta}_t$ vs. η/β on a log-log axis. The theorem predicts a slope of exactly 1. A slope significantly > 1 falsifies the bound (constant grows with η); a slope significantly < 1 suggests an additional saturating mechanism (e.g. the rank-2.2 plateau is structural rather than dynamical) and would itself be a strong finding.